The gap between the two walls is positive: a closed form for the (Ind2)-terms of lana-agents/iut

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Abstract

The public Lean formalization lana-agents/iut writes the comparison step in the proof of Corollary 3.12 of inter-universal Teichmüller theory (IUT) III—the "wall"—in two forms: a variant of Corollary 3.12, (Var), and the Scholze–Stix inequality, (SS). The difference $\Delta(D)$ of their right-hand sides is a weighted average of (Ind2)-terms at the bad residue characteristics. The repository proves $\Delta(D) \geq 0$ and records as open whether these terms are zero or positive.

We determine these terms in closed form. The (Ind2)-term equals $(\mathrm{ord}_p\,\theta - \kappa_v(\theta))\log p$, where $\theta = \underline{q}_v^{j^2}$ is the theta value and $\kappa_v(\theta)$ is the largest integer k with $\theta\,\mathcal{I}_v \subseteq p^k\mathcal{I}_v$, $\mathcal{I}_v$ being the log-shell. Consequently the term is always at least $\mathrm{frac}(\mathrm{ord}_p\,\theta)\log p$, with equality when p is odd and $e_v \leq p-2$ (there the log-shell is a fractional ideal). At a bad place $\mathrm{ord}_p\,\theta$ is never an integer, because ℓ does not divide the order of q. Hence $\Delta(D) > 0$ for every initial Θ-datum, and in this formalization (SS) is strictly stronger than (Var).

On the other hand $\Delta(D)$ is bounded above by a quantity that does not involve the size of the q-orders (the values $\log p$ at the bad places and the ramification indices). The two walls are height inequalities of the same shape that differ by a term of conductor type. The average over the labels is given by the class number of $\mathbb{Q}(\sqrt{-\ell})$. We compute the values for six explicit initial Θ-data.

No claim is made about the truth of (Var), (SS), Corollary 3.12, or the abc conjecture.

Contents

1 Introduction **2**
1.1 Motivation . . . 2
1.2 The two walls and their gap . . . 2
1.3 Results . . . 3
1.4 Relation to earlier work . . . 3
1.5 What is not claimed . . . 3
1.6 Labels . . . 4

2 Setting **4**

3 Main results **5**

4 Proofs **6**
4.1 The lattice lemmas . . . 6
4.2 Proof of Theorem 3.2 . . . 6
4.3 Proof of Theorem 3.3 . . . 7

4.4 Proof of Theorem 3.4 7
4.5 Proofs of Corollary 3.5 and Proposition 3.6 7

5 Examples **7**

6 Discussion **8**
6.1 How the two walls relate 8
6.2 Readings of the (Ind2) group 8
6.3 Comparison with the estimate of IUT IV 9

7 Reproducibility **9**

1 Introduction

1.1 Motivation

The ultimate aim of this work is to understand exactly the algorithm by which IUT computes the height inequality. The computation of IUT IV combines, step by step, theta values, log-shells, the indeterminacies (Ind1), (Ind2), (Ind3), holomorphic hulls and log-volumes, and produces the two sides of the height inequality. To write this algorithm down exactly, including the value produced at each step, every step must have a determined value. The (Ind2)-term, whose value the public formalization lana-agents/iut leaves open, is one of these steps. This note determines it.

1.2 The two walls and their gap

The repository lana-agents/iut [lana] ("lana" below) is a public Lean formalization that builds the objects of IUT. Its directory `Iut/WallEquiv/` collects formulations of the step from Theorem 3.11 to Corollary 3.12 of IUT III [IUT3], the "wall" in the words of the interim report of Project LANA [LANA, §0.2], and proves equivalences between them in its models. The file `Gap.lean` treats the difference between two of them.

> X "Christian's question: *is the LANA wall any weaker than the Scholze–Stix wall?*" "**Open**: whether the (Ind2)-terms at the bad primes are `0` or `> 0` (the indeterminacy automorphisms at a prime are all $\mathbb{Z}_p$-lattice automorphisms of the factor log-shells, `tensorAut`, which need not commute with the theta values); so the sign Δ`(D) ≥ 0` is all that is proved."
>
> [lana], `Iut/WallEquiv/Gap.lean`, module docstring

For initial Θ-data D the two walls are the following inequalities (notation in Section 2).

(Var) $-|\log q| \le R$, where $R = -|\log \Theta|$ is the log-volume of the holomorphic hull of the Θ-pilot region: a variant of Corollary 3.12.

(SS) $-|\log q| \le \mu^{\log}(\mathrm{LGP} \cdot \mathrm{hull}(VC(I)))$: the Scholze–Stix inequality [SS], written with IUT's container $\mathrm{hull}(VC(I))$.

lana defines the gap $\Delta(D) = R - \mu^{\log}(\mathrm{LGP} \cdot \mathrm{hull}(VC(I))) \ge 0$, proves that (SS) is equivalent to $-|\log q| \le R - \Delta(D)$ (`ss_iff_le_rhs_sub_gap`), and proves that $\Delta(D)$ is a weighted average of the (Ind2)-terms at the bad residue characteristics (`gap_eq`). The same repository expects positivity:

> X "they do not commute with the multiplication by `q^{j^2}` in general, so `ind2Term > 0` is to be expected when `ord(q^{j^2})` is not a multiple of the ramification index, but this is not formalized."
>
> [lana], `Iut/WallEquiv/README.md`, section "The gap between (Var) and (SS) (`Gap.lean`)"

1.3 Results

Working with lana's definitions as they stand, we determine the (Ind2)-terms.

- Theorem 3.2 (closed form): $\mathrm{ind2} = (\mathrm{ord}_p\,\theta - \kappa_v(\theta))\log p$.
- Theorem 3.3 (bounds): $\mathrm{ind2} \geq \mathrm{frac}(\mathrm{ord}_p\,\theta)\log p$ and $\mathrm{ind2} < (1 + a_v + b_v)\log p$, with equality in the lower bound when p is odd and $e_v \leq p-2$; here a_v, b_v are the constants of Proposition 1.2 of IUT IV [IUT4].
- Theorem 3.4 (positivity): $\mathrm{ind2} > 0$ at every bad place. Hence $\Delta(D) > 0$ for every initial Θ-datum, and (SS) is strictly stronger than (Var).
- Corollary 3.5 (size): $\Delta(D) < \sum_u w_u(\log p_u + 3 + \log e_u)$; the right-hand side does not involve the size of the q-orders.
- Proposition 3.6 (label average): if $\mathrm{ord}_p(q_v) = n/\ell$, the average of $\mathrm{frac}(\mathrm{ord}_p\,\theta)$ over the labels is $\frac{1}{2} - \chi_\ell(n)\,h(-\ell)/(\ell-1)$ when $\ell \equiv 3 \bmod 4$; at small places this is the average of the term in units of $\log p$.

Our answer to lana's question is therefore: in lana's formalization the LANA wall is strictly weaker than the Scholze–Stix wall, but only by a quantity of conductor type, which does not grow with the size of q.

X The title of this note is close to the message of lana's commit `6ef57cb` (13 October 2026), "WallEquiv/Gap: the gap between (Var) and (SS) is positive for every D". That commit proved the positivity of the gap to the form of (SS) without the hull, from the archimedean defect (`gap_pos`); since the switch to the hull in commit `cdc9602` (14 October 2026) this theorem is `gapNoHull_pos`. With the hull on both sides these defects cancel ("The shell and archimedean defects are exactly what the hull removes", `Gap.lean`), and for $\Delta(D)$ the file proves the sign $\Delta(D) \geq 0$ and leaves positivity open, as quoted above. The positivity in the title of this note concerns $\Delta(D)$.

1.4 Relation to earlier work

- X Positivity itself is expected, under a condition, in lana's documentation quoted above. We give a proof and the exact value.
- X Dupuy and Hilado describe (Ind2) as acting through the $\mathbb{Q}_p$-linear automorphisms arising from $\mathbb{Z}_p$-lattice automorphisms of $\mathcal{I}_v$ [DH1, §4.8], and bound the hull under such automorphisms *from above* [DH2, §6.2]; the floor $\lfloor \mathrm{ord}_p(a) - \mathrm{ord}_p(\beta) \rfloor$ appears in their estimate (6.6)–(6.7). Theorem 3.2 shows that the hull actually reaches the position of this rounding. As an alternative that they do not pursue, they suggest writing out an explicit $\mathbb{Z}_p$-basis and computing the action of these automorphisms explicitly [DH2, Rem. 6.2.1(1)]; Lemmas 4.1–4.3 compute, for the lattices $\theta \cdot \bigotimes \mathcal{I}$ of lana's definitions, the span of their images under these automorphisms.
- S Proposition 1.2(ii) of IUT IV [IUT4] likewise bounds $\phi(p^\lambda(R_I)^\sim)$ from above by inclusions involving a floor, and (iii) replaces the floor by an allowance of 1. Neither states an equality.
- P The lattice lemma (Lemma 4.1) is elementary and not new. What is new, to the author's knowledge, is its application to lana's definitions to determine the (Ind2)-terms, and the positivity, bounds and label average that follow.

1.5 What is not claimed

- The object of this note is the definitions of lana's formalization. lana states that its (Ind2) group is a "deliberate enlargement" that contains the image of Ism (Section 2). For the

group of the original papers the value lies between 0 and the value found here; under the identification of [Fourth, Prop. 9] it is 0 (Section 6.2).

- Nothing is claimed about the truth of (Var), (SS), Corollary 3.12 or the abc conjecture.
- The proofs are on paper. They have not been checked in Lean on top of lana's definitions.

1.6 Labels

S a statement of the original papers of IUT or of Mochizuki's documents. X a statement of others (lana, Project LANA, Dupuy–Hilado, Zeng). P a statement proved here; P alone marks an elementary fact. R known mathematics, applied here in a new way; the known mathematics is named. N a result new in this note, to the author's knowledge. H a reading or interpretation, not a proof.

2 Setting

We follow lana's definitions; Lean names are given in parentheses.

- *Initial* Θ*-data* D: a number field F, an elliptic curve E/F, a prime $\ell \geq 5$, $\ell^* = (\ell-1)/2$, $K = F(E[\ell])$, the field of moduli F_{mod} and a nonempty set $\mathbb{V}_{\mathrm{mod}}^{\mathrm{bad}}$ of bad places (`bad_nonempty`). At a bad place w the order of the Tate parameter q_w is prime to ℓ (`q_order_coprime`; IUT I, Definition 3.1(c) [IUT1]).
- *Capsules and labels*: the labels of capsule $i = 0, \dots, \ell^* - 1$ are $\{0, \dots, i+1\}$, and the distinguished label is $j = i+1$ (`distinguished`). Thus j runs through $1, \dots, \ell^*$.
- *Packets*: for a prime p and a map (tuple) c from the labels to the places of F_{mod} over p, $T_c = \bigotimes_l K_{v_l}$, the tensor product over $\mathbb{Q}_p$, where v_l is the place of K over $c(l)$ given by the section of places.
- *Log-shells*: $\mathcal{I}_v = (2p)^{-1} \log_p(\mathcal{O}_v^\times) \subseteq K_v$, a $\mathbb{Z}_p$-lattice (`factorLogShell`); $\bigotimes \mathcal{I}$ is the $\mathbb{Z}_p$-submodule spanned by the pure tensors of log-shell elements (`LocalTheory.shell` = `tensorLogShell`).
- *Integral structure and hull*: $\mathcal{O}$ is the integral closure of $\mathbb{Z}_p$ in T_c (`integral`). The hull $\mathrm{hull}(U)$ of a subset U is the intersection of all $b\,\mathcal{O}$, b a unit, that contain U (`leastHull`; IUT III, Remark 3.9.5(i) [IUT3]).
- *Log-volume*: $\mu^{\log}(S) = \log(\mu(S)/\mu(\mathcal{O}))/[T_c : \mathbb{Q}_p]$ (`componentVol`). Multiplication by p shifts it by $-\log p$, and multiplication by y in the factor of label l by $-\mathrm{ord}_p(y)\log p$ (`componentVol_prod_incl_smul_integral`), with ord_p normalized by $\mathrm{ord}_p(p) = 1$.
- *The (Ind2) group*: tensorAut consists of the tensor products $\bigotimes_l g_l$ of $\mathbb{Q}_p$-linear automorphisms g_l of the factors with $g_l(\mathcal{I}_{v_l}) = \mathcal{I}_{v_l}$ (`tensorAut`).
- *Theta values*: $\underline{q}_v$ is a 2ℓ-th root of the Tate parameter at a bad place v, and 1 elsewhere (`qrootOf`); $\theta = \underline{q}_v^{\,j^2}$, with $v = v_j$, sits in the factor of label j (`scaleElt`). One has $\mathrm{ord}_p(\underline{q}_v) = \mathrm{ord}_w(q_w)/(2\ell\, e_w)$, where w is the place of F below v and e_w the ramification index of $F_w/\mathbb{Q}_p$ (`ordp_qrootOf`).
- *The* Θ*-pilot component* (in IUT IV's reading): $\Theta_{i,p,c} = \bigcup_{\phi \in \mathrm{tensorAut}} \phi(\theta \cdot \bigotimes \mathcal{I})$ (`thetaFinite_eq`).
- *The (Ind2)-term* (`ind2Term`):

$$\mathrm{ind2}(i,p,c) = \mu^{\log}(\mathrm{hull}(\Theta_{i,p,c})) - \mu^{\log}(\theta \cdot \mathrm{hull}(\textstyle\bigotimes \mathcal{I})).$$

- *The gap* (`gap_eq`):
$$\Delta(D) = \frac{1}{\ell^*} \sum_i \sum_p \sum_c w_c \,\mathrm{ind2}(i,p,c),$$
with $w_c = \prod_l w(c(l))$ and $w(u) = [F_{\mathrm{mod},u} : \mathbb{Q}_p]/[F_{\mathrm{mod}} : \mathbb{Q}]$. The weights of the places over p sum to 1 (`weight_sum_one`).

> X "This class contains the image of `Ism` and is in general larger: a **deliberate enlargement**, which makes the theta region larger and the hypothesis `Cor312VariantHolds` weaker than with `Ism` exactly."
>
> [lana], `Iut/Concrete/LocalConstruct/IndAut.lean`, module docstring

We use the constants of IUT IV, Proposition 1.2 [IUT4], for the ramification index e of $K_v/\mathbb{Q}_p$:
$$a = \tfrac{1}{e}\lceil \tfrac{e}{p-2} \rceil \ (p>2), \quad a = 2 \ (p=2), \qquad b = \Big\lfloor \frac{\log(pe/(p-1))}{\log p} \Big\rfloor - \frac{1}{e}.$$
By Proposition 1.2(i) of IUT IV, $p^a \mathcal{O}_v \subseteq \log_p(\mathcal{O}_v^\times) \subseteq p^{-b}\mathcal{O}_v$, with equality in both inclusions when $p > 2$ and $e \le p-2$; then $a = 1/e = -b$.

Definition 2.1. A finite place $v \mid p$ of K is *small* if p is odd and $e_v \le p-2$. At a small place $\log_p(\mathcal{O}_v^\times) = p^{1/e_v}\mathcal{O}_v$, so $\mathcal{I}_v$ is an $\mathcal{O}_v$-module.

3 Main results

Definition 3.1. For a finite place $v \mid p$ of K and $\theta \in K_v^\times$ put
$$\kappa_v(\theta) = \max\{k \in \mathbb{Z} : \ \theta\,\mathcal{I}_v \subseteq p^k\,\mathcal{I}_v\}.$$
The maximum exists because $\theta\,\mathcal{I}_v$ is again a $\mathbb{Z}_p$-lattice.

Theorem 3.2 (closed form). P N *Let $v = v_j$ and $\theta = \underline{q}_v^{\,j^2}$. Then*
$$\mathrm{ind2}(i,p,c) = \big(\mathrm{ord}_p\,\theta - \kappa_v(\theta)\big)\log p.$$
In particular the term depends only on j and v_j, not on the places at the other labels.

Theorem 3.3 (bounds). P N *With the same notation and e_v the ramification index of $K_v/\mathbb{Q}_p$,*
$$\mathrm{frac}(\mathrm{ord}_p\,\theta)\,\log p \ \le\ \mathrm{ind2}(i,p,c) \ \le\ \big(\mathrm{ord}_p\,\theta - \lfloor \mathrm{ord}_p\,\theta - a_v - b_v \rfloor\big)\log p \ <\ (1 + a_v + b_v)\log p.$$
If v is small then $a_v + b_v = 0$, the two bounds coincide, and $\mathrm{ind2}(i,p,c) = \mathrm{frac}(\mathrm{ord}_p\,\theta)\log p$.

Theorem 3.4 (positivity). P N

(i) If v_j is a bad place, then $\mathrm{ord}_p\,\theta \notin \mathbb{Z}$ *for every $j = 1, \dots, \ell^*$, and* $\mathrm{ind2}(i,p,c) > 0$.

(ii) $\Delta(D) > 0$ *for every initial Θ-datum D. Hence in lana's formalization (SS) is strictly stronger than (Var): (Var) holds and (SS) fails exactly when $-|\log q| \in (R - \Delta(D), R]$ (`variant_and_not_ss_iff`), and this interval has positive length $\Delta(D)$.*

Corollary 3.5 (formula and size of the gap). P *For $u \in \mathbb{V}_{\mathrm{mod}}^{\mathrm{bad}}$ let $v(u)$ be the place of K over u, $r_u = \mathrm{ord}_p(\underline{q}_{v(u)})$, and p_u the residue characteristic. Then*
$$\Delta(D) = \frac{1}{\ell^*} \sum_{j=1}^{\ell^*} \sum_{u \in \mathbb{V}_{\mathrm{mod}}^{\mathrm{bad}}} w_u \,\big(j^2 r_u - \kappa_{v(u)}(\underline{q}_{v(u)}^{\,j^2})\big)\log p_u,$$

$$\frac{1}{\ell^*}\sum_{j=1}^{\ell^*}\sum_{u\in\mathbb{V}_{\mathrm{mod}}^{\mathrm{bad}}} w_u\,\mathrm{frac}(j^2 r_u)\log p_u \;\le\; \Delta(D) \;<\; \sum_{u\in\mathbb{V}_{\mathrm{mod}}^{\mathrm{bad}}} w_u(\log p_u + 3 + \log e_{v(u)}).$$

If every bad place is small, the lower bound is an equality. The upper bound does not involve the size of the q-orders.

Proposition 3.6 (label average and class number). P R

Known mathematics: Dirichlet's class number formula.

Suppose $r_u = n_u/\ell$ with $n_u \in \mathbb{Z}$ and $\ell \nmid n_u$. Then

$$\frac{1}{\ell^*}\sum_{j=1}^{\ell^*}\mathrm{frac}\Big(\frac{j^2 n_u}{\ell}\Big) = \begin{cases} \dfrac{1}{2} - \chi_\ell(n_u)\,\dfrac{h(-\ell)}{\ell-1} & (\ell \equiv 3 \bmod 4),\\[2mm] \dfrac{1}{2} & (\ell \equiv 1 \bmod 4), \end{cases}$$

where χ_ℓ is the Legendre symbol and $h(-\ell)$ the class number of $\mathbb{Q}(\sqrt{-\ell})$. For $\ell = 7$ one has $h(-7) = 1$, so the average is $1/3$ if n_u is a quadratic residue and $2/3$ otherwise.

4 Proofs

4.1 The lattice lemmas

Lemma 4.1. P *Let L be a free $\mathbb{Z}_p$-module of finite rank, $M \subseteq L \otimes \mathbb{Q}_p$ a $\mathbb{Z}_p$-lattice, and $\kappa = \max\{k : M \subseteq p^k L\}$. Then the $\mathbb{Z}_p$-submodule spanned by $\bigcup_{g\in\mathrm{GL}(L)} g(M)$ is $p^\kappa L$.*

Proof. Since $g(M) \subseteq g(p^\kappa L) = p^\kappa L$, the span lies in $p^\kappa L$. Conversely, by maximality of κ there is $x \in M \setminus p^{\kappa+1}L$, and $y = p^{-\kappa}x$ lies in $L \setminus pL$. By Nakayama's lemma y extends to a basis of L, so for every vector e of a basis of L there is $g \in \mathrm{GL}(L)$ with $g(y) = e$. Then $p^\kappa e = g(x) \in g(M)$. The vectors $p^\kappa e$ span $p^\kappa L$. □

Lemma 4.2. P *For $U \subseteq T_c$ and a unit b, $U \subseteq b\,\mathcal{O}$ if and only if $\mathrm{span}_{\mathbb{Z}_p} U \subseteq b\,\mathcal{O}$. Hence $\mathrm{hull}(U) = \mathrm{hull}(\mathrm{span}_{\mathbb{Z}_p} U)$.*

Proof. $b\,\mathcal{O}$ is a $\mathbb{Z}_p$-submodule. □

Lemma 4.3. P *The $\mathbb{Z}_p$-submodule spanned by $\Theta_{i,p,c}$ is $p^\kappa \bigotimes \mathcal{I}$, where $\kappa = \kappa_{v_j}(\theta)$.*

Proof. Since θ acts on the factor of label j, $\theta \cdot \bigotimes \mathcal{I} = \mathcal{I}_{v_1} \otimes \cdots \otimes \theta\mathcal{I}_{v_j} \otimes \cdots$. For $\phi = \bigotimes_l g_l \in \mathrm{tensorAut}$ one has $g_l(\mathcal{I}_{v_l}) = \mathcal{I}_{v_l}$, so $\phi(\theta \cdot \bigotimes \mathcal{I}) = \mathcal{I}_{v_1} \otimes \cdots \otimes g_j(\theta\mathcal{I}_{v_j}) \otimes \cdots$. Writing A for the free module formed by the other factors, the span of the union of these images is $A \otimes \mathrm{span}(\bigcup_{g_j} g_j(\theta\mathcal{I}_{v_j}))$. By Lemma 4.1 with $L = \mathcal{I}_{v_j}$ and $M = \theta\mathcal{I}_{v_j}$, this is $A \otimes p^\kappa \mathcal{I}_{v_j} = p^\kappa \bigotimes \mathcal{I}$. □

4.2 Proof of Theorem 3.2

By Lemmas 4.2 and 4.3, $\mathrm{hull}(\Theta_{i,p,c}) = \mathrm{hull}(p^\kappa \bigotimes \mathcal{I})$. The hull commutes with multiplication by units (lana, `leastHull_units_smul`), so this is $p^\kappa\,\mathrm{hull}(\bigotimes \mathcal{I})$. Put $X = \mathrm{hull}(\bigotimes \mathcal{I})$. Log-volumes of unit multiples shift additively (`componentVol_units_smul`): multiplication by p^κ shifts by $-\kappa \log p$, and multiplication by θ by $-\mathrm{ord}_p(\theta)\log p$ (`componentVol_prod_incl_smul_integral`). Hence

$$\mathrm{ind2} = \big(\mu^{\log}(X) - \kappa\log p\big) - \big(\mu^{\log}(X) - \mathrm{ord}_p(\theta)\log p\big) = (\mathrm{ord}_p\,\theta - \kappa)\log p.$$

4.3 Proof of Theorem 3.3

Lower bound. Compare the Haar measures of the two sides of $\theta\mathcal{I}_v \subseteq p^{\kappa}\mathcal{I}_v$ in K_v. With $n = [K_v : \mathbb{Q}_p]$, multiplication by θ multiplies measures by $p^{-n \operatorname{ord}_p \theta}$ and multiplication by p^{κ} by $p^{-n\kappa}$, so $\kappa \le \operatorname{ord}_p \theta$. As κ is an integer, $\kappa \le \lfloor \operatorname{ord}_p \theta \rfloor$, and Theorem 3.2 gives

$$\mathrm{ind2} \ge (\operatorname{ord}_p \theta - \lfloor \operatorname{ord}_p \theta \rfloor) \log p = \operatorname{frac}(\operatorname{ord}_p \theta) \log p.$$

Upper bound. IUT IV, Proposition 1.2(i), gives $\theta\mathcal{I}_v \subseteq p^{\lfloor \operatorname{ord}_p \theta - a_v - b_v \rfloor}\mathcal{I}_v$ (lana, `exists_mul_mem_logShell`). Hence $\kappa \ge \lfloor \operatorname{ord}_p \theta - a_v - b_v \rfloor$, and Theorem 3.2 gives the upper bound; $x - \lfloor x - c \rfloor < 1 + c$ gives the last inequality.

Small places. Here $a_v + b_v = 0$ (IUT IV, Proposition 1.2; lana, `logLatticeA_add_B_eq_zero`), so the two bounds coincide. □

4.4 Proof of Theorem 3.4

(i) $\operatorname{ord}_p \theta = j^2 \operatorname{ord}_p(\underline{q}_v) = j^2 \operatorname{ord}_w(q_w)/(2\ell e_w)$. If this were an integer m, then $j^2 \operatorname{ord}_w(q_w) = 2\ell e_w m$, and the prime ℓ would divide $j^2 \operatorname{ord}_w(q_w)$. But $1 \le j \le \ell^* < \ell$, so $\ell \nmid j$, and $\ell \nmid \operatorname{ord}_w(q_w)$ (`q_order_coprime`). Hence $\operatorname{frac}(\operatorname{ord}_p \theta) > 0$, and $\mathrm{ind2} > 0$ by Theorem 3.3.

(ii) Each term of the formula for $\Delta(D)$ is the product of $w_c > 0$ (`weight_pos`) and $\mathrm{ind2} \ge 0$ (`ind2Term_nonneg`). Since $\mathbb{V}^{\mathrm{bad}}_{\mathrm{mod}}$ is nonempty, take $u \in \mathbb{V}^{\mathrm{bad}}_{\mathrm{mod}}$, let p be its residue characteristic and c a tuple with $c(j) = u$; by (i) that term is positive. □

4.5 Proofs of Corollary 3.5 and Proposition 3.6

Corollary 3.5. By Theorem 3.2 the term depends only on $c(j)$. Since the weights over p sum to 1, $\sum_c w_c\, f(c(j)) = \sum_u w_u\, f(u)$. At places that are not bad, $\underline{q} = 1$ and the term is 0. This gives the formula; the lower bound and the case of equality follow from Theorem 3.3. The upper bound follows from the bound $(1 + a_v + b_v)\log p$ of Theorem 3.3 and from $(a_v + b_v)\log p \le 3 + \log e_v$ (lana, `logLatticeA_add_B_le`). □

Proposition 3.6. As j runs through $1, \dots, \ell^*$, $j^2 \bmod \ell$ runs through the set Q of quadratic residues, each once. The sum is therefore $\frac{1}{\ell}\sum_{x \in nQ} x$ (representatives in $1, \dots, \ell - 1$), and nQ is Q if n is a residue and the set N of non-residues otherwise. If $\ell \equiv 1 \bmod 4$, then $-1 \in Q$ and $\sum_Q x = \ell(\ell-1)/4$. If $\ell \equiv 3 \bmod 4$ and $\ell > 3$, Dirichlet's class number formula $h(-\ell) = \frac{1}{\ell}(\sum_N x - \sum_Q x)$ and $\sum_Q + \sum_N = \ell(\ell-1)/2$ give $\sum_Q x = \ell(\ell-1)/4 - \ell h(-\ell)/2$. Divide by ℓ^*. □

5 Examples

P We computed the six explicit initial Θ-data with $\ell = 7$ of the author's catalogue [Cat] ($\ell^* = 3$, $s = \ell(\ell+1)/12 = 14/3$); its seventh datum, $D^{(\ell)}_{2357}$, has $\ell \ge 89$ and is not treated here. The catalogue takes as $\mathbb{V}^{\mathrm{bad}}_{\mathrm{mod}}$ the places of multiplicative reduction outside 2ℓ. We take this set minus the places above 3 and 5, since above 3 and 5 the ramification of K depends on the construction of F; in these data no other place is left out, and $\ell \nmid \operatorname{ord} q$ at every place we keep. IUT I, Definition 3.1(b), requires $\mathbb{V}^{\mathrm{bad}}_{\mathrm{mod}}$ only to be a nonempty set of nonarchimedean places of odd residue characteristic such that X_F has bad (multiplicative) reduction at the places of F over it, and notes that X_F may have bad reduction over $\mathbb{V}^{\mathrm{good}}_{\mathrm{mod}}$ as well; Definition 3.1(c) requires ℓ to be prime to the residue characteristics of $\mathbb{V}^{\mathrm{bad}}_{\mathrm{mod}}$ and to the q-orders there. So this choice is allowed [IUT1]. The lower bound is the sum of fractional parts in Theorem 3.3 and depends only on the data. The upper bound uses the constants a, b of IUT IV, Proposition 1.2. For the ramification index of $K_v/\mathbb{Q}_p$ we take $e(F_{\mathrm{mod},u}/\mathbb{Q}_p) \cdot 15\ell/\gcd(15\ell, \operatorname{ord}_u q)$; above the special prime this is the formula of Appendix A of the catalogue, at the other places it is an assumption. The assumption affects only which places are small and the upper bound.

datum	bad places (small, not small)	lower bound of $\Delta(D)$	upper bound of $\Delta(D)$	$(s-1)\lvert\log q\rvert$	ratio
D211	4, 1	25.894	27.933	79.937	0.324–0.349
D109	6, 0	22.707	22.707 (exact)	70.085	0.324
D2	3, 2	28.516	30.914	62.881	0.453–0.492
D3	14, 5	93.174	102.691	295.492	0.315–0.348
D18	14, 3	53.621	58.621	161.059	0.333–0.364
D15	13, 2	45.765	49.165	135.160	0.339–0.364

Example 5.1 (D109). P There are six bad places, with residue characteristics 109, 21577, 6065177, 534775081, 1174770007439769673 and 16450055877403724897. All weights are $1/2$, and r_u is $29/7$ at 109 and $1/7$ at the others. Even at the smallest, $p = 109$, one has $e_v = 105 \le p-2$ (by the formula of Appendix A of the catalogue), so all six places are small. Both $n_u = 29$ and $n_u = 1$ are quadratic residues modulo 7, so by Proposition 3.6 the label average is $1/3$ and

$$\Delta(D_{109}) = \sum_u \frac{1}{2}\cdot\frac{1}{3}\log p_u = \frac{1}{6}\sum_u \log p_u = 22.7068\ldots$$

The ratio to $(s-1)\lvert\log q\rvert = 70.085$ is 0.324.

Example 5.2 (a model of a place that is not small). P Let $K_v = \mathbb{Q}_3(\varpi)$ with $\varpi^4 = 3$ ($p = 3$, $e = 4$). We computed $\mathcal{I}_v$ and κ exactly. For $\theta = \varpi$, $\operatorname{ord}_p\theta = 1/4$ and $\kappa = -1$, so the term is $\frac{5}{4}\log 3$, which exceeds $\log p$. For $\theta = \varpi^4$, $\kappa = 1$ and the term is 0; for $\theta = (1+\varpi)\varpi^4$, $\kappa = 0$ and the term is $\log 3$: the value depends on the unit part. All three lie inside the upper bound $(\operatorname{ord}_p\theta - \lfloor\operatorname{ord}_p\theta - a - b\rfloor)\log p$ of Theorem 3.3, here with $a+b = 7/4$. This model is for illustration only and is not a bad place of actual data (at a bad place $\operatorname{ord}_p\theta \notin \mathbb{Z}$).

H In these data the q-order is 2 at most bad places, which is small, so $\Delta(D)$ is between three and five tenths of the main term $(s-1)\lvert\log q\rvert$. That $\Delta(D)$ is small holds only in the direction of growing q-orders; its actual size is set by the values $\log p$ at the bad places.

6 Discussion

6.1 How the two walls relate

P With lana's identity $\Delta(D) = R + s\lvert\log q\rvert - V$, where $V = \mu^{\log}(\mathrm{hull}(VC(I)))$ (`gap_eq_rhs_add`), the two walls read

$$\text{(SS)}\quad (s-1)\lvert\log q\rvert \le V, \qquad\qquad \text{(Var)}\quad (s-1)\lvert\log q\rvert \le V + \Delta(D).$$

H V is bounded above in terms of ramification and does not involve the q-orders (lana, `Iut/WallEquiv/VcI.lean`); by Corollary 3.5, $\Delta(D)$ is a quantity of the same kind. So the two walls are height inequalities of the same shape that differ only in a term of conductor type on the right-hand side. The difference is positive (Theorem 3.4).

6.2 Readings of the (Ind2) group

- P IUT IV, Proposition 1.2, considers arbitrary $\mathbb{Q}_p$-linear automorphisms ϕ preserving $\bigotimes\mathcal{I}$, a larger group than tensorAut. The value does not change for that group: since θ acts on one factor only, $\theta\cdot\bigotimes\mathcal{I} \subseteq p^k\bigotimes\mathcal{I}$ if and only if $\theta\mathcal{I}_{v_j} \subseteq p^k\mathcal{I}_{v_j}$, so Lemma 4.1 applied to $L = \bigotimes\mathcal{I}$ yields the same κ.

- P For a smaller group $G \subseteq \mathrm{tensorAut}$ the union only shrinks, so the term lies between 0 and the value found here. The image of the group Ism of the original papers is such a case (lana's documentation).
- P At a small place, if G consists only of multiplications by $\mathbb{Z}_p^\times$ and field automorphisms, the term is 0, because $\sigma(\theta)\mathcal{I}_v = \theta\mathcal{I}_v$. If moreover every bad place is small, then $\Delta(D) = 0$ and (SS) and (Var) coincide.
- H Where the value for the group of the original papers lies in this range depends on how far that group moves $\theta\mathcal{I}_v$. The proof of Lemma 4.1 uses elements that send a primitive element of $\theta\mathcal{I}_v$ to a basis vector. Whether the image of Ism contains such elements is not decided here.
- P The author's fourth report [Fourth, §6, Proposition 9] applies [MT, Thm. F, Thm. 7.4] to a finite extension K_v of $\mathbb{Q}_p$: there $\mathrm{Ism}(G_{K_v}) = \mathbb{Z}_p^\times$, so (Ind2) acts on each factor by a unit scalar. If the group of the original papers is identified with Ism in this way, it fixes every $\theta\mathcal{I}_v$, since $\mathcal{I}_v$ is a $\mathbb{Z}_p$-module; the union $\Theta_{i,p,c}$ is then $\theta \cdot \bigotimes \mathcal{I}$, and its (Ind2)-term is 0, because $\mathrm{hull}(\theta X) = \theta\ \mathrm{hull}(X)$ (proof of Theorem 3.2). For that group $\Delta(D) = 0$, and (SS) and (Var) coincide.

6.3 Comparison with the estimate of IUT IV

S IUT IV, Proposition 1.2(iii) states: if $p > 2$ and $e_i \le p-2$ for all $i \in I$, then $\phi(p^\lambda \cdot (R_I)^\sim) \subseteq p^{\lambda - \mathfrak{d}_I - 1} \cdot (R_I)^\sim$ for every $\lambda \in \frac{1}{e_i}\mathbb{Z}$, $i \in I$ [IUT4].

H Besides the different term $\mathfrak{d}_I$, this estimate allows 1 for rounding. The term $\mathfrak{d}_I$ comes from comparing $\bigotimes_i R_i$ with its normalization $(R_I)^\sim$ (the proof of (ii) uses Proposition 1.1 for this). In lana's $\Delta(D)$ the hull is taken on both sides, so this comparison, the shell defect, cancels (the docstring of `gap_eq`; for the shell defect see `Iut/WallEquiv/VcI.md`), and only the rounding part remains. Theorem 3.3 shows that under lana's group this rounding loss actually occurs, and at small places its size is exactly $\mathrm{frac}(\mathrm{ord}_p\,\theta) \in (0,1)$ in units of $\log p$. Averaged over the labels it is about $1/2$ by Proposition 3.6 ($1/3$ or $2/3$ for $\ell = 7$). Compared with the 1 that the estimate of IV allows for rounding, the actual loss is about half of it on average over the labels.

P The unused part of this allowance, $1 - \mathrm{frac}(\mathrm{ord}_p\,\theta) = 1 + \lfloor \mathrm{ord}_p\,\theta \rfloor - \mathrm{ord}_p\,\theta$ at small places, is the floor slack that the author's third report computes on explicit data [Third, §6].

X Zeng [Zeng] (first version 29 October 2026) computes the index by which the hull enlarges the integral structure of a tensor product of rings of integers. P In $\Delta(D)$ both terms are log-volumes of unit multiples of the same hull $X = \mathrm{hull}(\bigotimes \mathcal{I})$ (proof of Theorem 3.2), so the enlargement of $\bigotimes \mathcal{I}$ to its hull, of the kind Zeng measures, cancels in $\Delta(D)$.

7 Reproducibility

The folder `ind2-gap/` of the repository `https://github.com/turahigashi/iut-explicit-initial-theta` (tag `v0.24.0`), also deposited with this note, contains the following scripts and their outputs. They use the program `iut_compute.py` of the author's walkthrough [Walk], which is in the same repository.

- `scripts/check_delta.py`: the table of Section 5 (lower bounds by fractional parts, upper bounds by the constants a, b), and Proposition 3.6 checked in exact rational arithmetic for twelve primes $\ell \equiv 3 \bmod 4$ ($7 \le \ell \le 83$) and six primes $\ell \equiv 1 \bmod 4$ ($5 \le \ell \le 41$), for all n.
- `scripts/delta2.py`: the same lower bounds, with upper bounds from the inner and outer radii of the log-shells.

- `scripts/shell_lattice.py`, `scripts/gl_check.py`: the log-shell $\mathcal{I}_v = (2p)^{-1}\log(1+\mathfrak{m})$ of $\mathbb{Q}_p(p^{1/e})$ in exact rational arithmetic, and the span of $\theta\mathcal{I}_v$ under random elements of $\mathrm{GL}(\mathcal{I}_v)$ compared with $p^{\kappa}\mathcal{I}_v$ (Lemma 4.1, Example 5.2) for $(p,e) = (11,3), (3,4), (5,6)$.

Acknowledgements and use of AI

The question studied in this note—to determine the value of the (Ind2)-terms that lana-agents/iut leaves open—arose in the author's discussions with generative AI tools; the author judged it important and decided to make it the subject of this note. Its ultimate aim, to understand exactly the algorithm by which IUT computes the height inequality, is the author's research direction. Generative AI tools (large language models) were used extensively, under the author's direction, in finding the proofs, in the computations and in the writing. The author takes full responsibility for the contents of this note.

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